

**SOUND-AWARE NOTIFICATION ASSISTANT (SANA): A
SMARTPHONE-BASED ENVIRONMENTAL SOUND
CLASSIFICATION AND ACOUSTIC EVENT
JOURNALING SYSTEM FOR THE
DEAF AND HARD-OF-HEARING**



Submitted in Partial Fulfillment Computer Science Thesis 1

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CHAPTER 1: INTRODUCTION

Background of the Study

Communication and environmental awareness are among the most fundamental human capacities, yet for an estimated 466 million individuals worldwide who experience disabling hearing loss, these capacities are mediated through alternative sensory and technological channels (World Health Organization [WHO], 2021). In the Philippines, the Philippine Statistics Authority (PSA, 2020) reported that hearing disability constitutes one of the most prevalent forms of disability, with an estimated 900,000 Filipinos identified as deaf or hard of hearing (DHH). Despite this significant demographic, assistive technology solutions designed for Filipino DHH users remain limited in number, scope, and local contextual relevance—particularly in applications that operate in real-time and on consumer-grade mobile hardware.

The proliferation of smartphones as ubiquitous computing platforms has opened substantial opportunities for deploying sophisticated assistive systems that run entirely on a user's personal device. Deep learning architectures such as YAMNet have demonstrated the capacity to classify environmental sounds with high accuracy on resource-constrained smartphone hardware, enabling fully on-device inference without reliance on cloud servers (Gougeh et al., 2024; Mohaimenuzzaman et al., 2022). Gougeh et al. (2024) developed ImmerseGuard—an on-device two-stage YAMNet-based sound event detection system for Android smartphones—achieving 90% real-time accuracy in alerting users to emergency sirens, vehicles, and household sounds. Similarly, Mohaimenuzzaman et al. (2022) demonstrated that deep learning-based audio classifiers can be compressed and deployed on resource-constrained hardware while maintaining near-state-of-the-art classification accuracy.

On-device sound alerting for DHH users has been explored from multiple angles. Kim et al. (2018) developed one of the earliest systematic home monitoring assistants for DHH users, utilizing bidirectional gated recurrent neural networks (BiGRU) for sound event detection and converting detected sounds into text display and haptic vibration. Asakura (2023) explored augmented-reality presentation of household sounds for DHH users, confirming through user surveys that visualizing surrounding acoustic environments using machine learning classifiers and spectrograms significantly enhanced situational awareness. Fanzeres et al. (n.d.) developed an on-device mobile environmental sound recognition application for DHH users, executing all processing locally—from audio feature extraction to classification and visual result presentation—and reported positive user feedback during field tests.

Sound alerting systems for DHH users must also address notification fatigue. Jain et al. (2020) introduced HomeSound, an in-home iterative sound awareness system for DHH users. Their three-week field deployment in four homes revealed that while participants reported increased awareness, constant watch vibrations caused by repetitive or misclassified alerts were poorly received and led to system disengagement. This critical finding underscores the necessity of intelligent alert

filtering and context-aware prioritization. Similarly, Jain et al. (2022) introduced ProtoSound, a personalized and scalable sound recognition system for DHH mobile users, demonstrating that user-driven customization of sound classifiers significantly improved system utility and reduced unwanted alert interruptions. Goodman et al. (2021) extended this by showing that automated sound recognition tools are most effective when users can customize which sound events they prioritize—directly informing the design of SANA's Context-Aware Sound Prioritization Algorithm.

Beyond real-time alerting, the behavioral documentation of a user's acoustic environment—referred to in the clinical audiology literature as Ecological Momentary Assessment (EMA)—has emerged as a valuable methodology for longitudinal understanding of how individuals interact with their sound environment. Andersson et al. (2023) investigated real-life benefit from hearing aid noise management, finding that smartphone-based EMAs combined with acoustic data logging provided more accurate and targeted insights than retrospective self-report questionnaires such as the SSQ12. Christensen et al. (2021) demonstrated that remotely logged longitudinal hearing aid usage data could be used to establish behavioral norms and associations with environmental factors. Jorgensen et al. (2025) further characterized how acoustic environment demand and diversity vary across time-of-day and days-of-week, finding that a single one-week sampling period provides a representative snapshot of a person's acoustic behavioral patterns. Crucially, none of these EMA methodologies have been applied to a dedicated DHH assistive mobile application—a gap that SANA directly addresses through its local SQLite Acoustic Event Journal (AEJ).

Chin et al. (2023) contributed a wearable edge computing device for DHH road safety using an EfficientNet-based ensemble model deployed on an Arduino Nano BLE Sense, classifying emergency sirens and vehicle sounds with haptic and OLED alerts—confirming the effectiveness of haptic-visual hybrid alerting in real-world road safety scenarios. Velmurugan & Prabhakar (n.d.) developed Vibe, an EdgeML-based sound indicator device trained on a locally collected Indian-context dataset, validating that locally trained models outperform generic pretrained models in culturally and acoustically specific environments—directly motivating SANA's fine-tuning of YAMNet for Philippine acoustic conditions.

Human-AI collaborative design of sound awareness systems has also been examined. Huang et al. (2024) introduced HACS, a novel framework in which AI models identify sounds by their acoustic characteristics, prompting DHH users to leverage contextual knowledge to resolve the specific event. This taxonomy-based approach directly informed SANA's context-aware environmental zone scoring. Veena & Aravindhar (2022) developed SCSHIP—a DNN-based urban sound classification system for hearing-impaired users—providing visual and haptic alerts, confirming the utility of deep learning-based classification for DHH assistive applications.

The Technology Acceptance Model (TAM), first formulated by Davis (1989), remains the most widely referenced framework for predicting user acceptance and system

adoption in information technology research. Davis (1989) validated two measurement scales—Perceived Usefulness and Perceived Ease of Use—as the primary determinants of user acceptance of computing systems, with reliabilities of 0.98 and 0.94 respectively. The System Usability Scale (SUS) further operationalizes usability measurement through a validated 10-item Likert-scale questionnaire. This study adopts both TAM and SUS frameworks to evaluate the usability and acceptance of SANA among its target user population.

SANA (Sound-Aware Notification Assistant) is a Flutter-based mobile application developed to address these intersecting gaps. The system integrates a pre-trained YAMNet TFLite model for on-device environmental sound classification (Gougeh et al., 2024; Mohaimenuzzaman et al., 2022), a local SQLite acoustic event journal that logs, stores, and surfaces behavioral insights from historical acoustic events (Andersson et al., 2023; Christensen et al., 2021; Jorgensen et al., 2025), and a suite of novel algorithmic contributions that distinguish SANA from prior work.

Three novel algorithmic contributions distinguish SANA from prior work:

1. A **Context-Aware Sound Prioritization Algorithm (CASPA)** that dynamically ranks sound alerts based on model confidence, temporal cooldowns, time of day, environmental context (GPS geofencing), and user-defined urgency—directly addressing the notification fatigue documented by Jain et al. (2020).
2. An **Acoustic Pattern Mining Algorithm (APMA)** that analyzes stored acoustic event journal logs to surface recurring behavioral patterns and environmental insights for the user—extending EMA principles (Andersson et al., 2023; Jorgensen et al., 2025) to a DHH mobile context.
3. A local **SQLite-based Acoustic Event Journal (AEJ)** that records, timestamps, and stores classified sound events with spatial and temporal metadata for longitudinal user review (Christensen et al., 2021).

Together, these components constitute a comprehensive, locally-contextualized, and privacy-preserving assistive platform that no prior system has unified in a single application.

Statement of the Problem

Despite the growing body of research in assistive technology for the DHH population and the technical maturity of smartphone-based sound classification systems, significant gaps persist in the availability of integrated, offline, and locally-contextualized assistive solutions for Filipino DHH users. The reviewed literature reveals that existing systems address isolated components of the assistive technology pipeline—environmental sound detection (Kim et al., 2018; Gougeh et al., 2024) or acoustic monitoring (Andersson et al., 2023)—but none unify these capabilities with context-aware prioritization and longitudinal acoustic journaling into a single, fully on-device mobile application.

Specifically, this study addresses the following problems:

1. **Lack of On-Device, Locally-Contextualized Sound Alerting:** Filipino DHH users lack access to a real-time sound classification and alerting system that operates entirely on-device and is optimized for Philippine acoustic environments. BridgeApp (n.d.) demonstrated local feasibility for mobile assistive tools, but lacked environmental sound classification and context-aware alert prioritization.
2. **Notification Fatigue from Competing Alerts:** No current system provides a unified, context-aware sound prioritization mechanism that incorporates temporal cooldowns, time-of-day context, environmental zone, and user-defined urgency simultaneously. Jain et al. (2020) explicitly identified constant vibration and misclassification-driven alert fatigue as critical barriers to DHH system adoption.
3. **Absence of Acoustic Event Journaling for DHH Empowerment:** Acoustic event journaling has not been implemented in any system designed specifically for DHH individuals. The EMA methodology (Andersson et al., 2023; Christensen et al., 2021; Jorgensen et al., 2025) has only been applied to hearing aid users in clinical settings.
4. **Lack of Behavioral Pattern Insights for DHH Users:** No existing DHH assistive mobile application provides users with actionable insights about their historical acoustic environments—identifying recurring sounds, peak-activity periods, and spatial patterns over time.

Statements of the Problems

1. What is the architectural design of the Sound-Aware Notification Assistant (SANA) in terms of its logical, physical (user interface), and functional components?
2. How does the system perform in terms of real-time environmental mediation and acoustic journaling?
3. How do users perceive the usability and acceptance of the SANA application based on the System Usability Scale (SUS) Questionnaire?

Objectives of the Study

General Objective

This study aims to develop and evaluate SANA: a Sound-Aware Notification Assistant—a smartphone-based application that utilizes the YAMNet model for on-device environmental sound classification with temporal and spatial logging, incorporating context-aware prioritization and acoustic pattern analysis for Filipino deaf and hard-of-hearing users.

Specific Objectives

Specifically, this study seeks to achieve the following objectives:

1. To develop the Sound-Aware Notification Assistant (SANA), a smartphone-based application that utilizes the YAMNet model for on-device environmental sound classification with temporal and spatial logging.
2. To analyze the users' behavior and perception towards the SANA application using the System Usability Scale (SUS) Questionnaire.
3. To describe the architectural design of the SANA application, including its Context-Aware Sound Prioritization Algorithm that dynamically ranks alerts based on time of day, environmental context, and user-defined urgency.
4. To design and implement a Context-Aware Sound Prioritization Algorithm (CASPA) that dynamically ranks classified acoustic alerts based on time of day, environmental context, and sound urgency to minimize notification fatigue (Jain et al., 2020) and maximize user safety awareness.
5. To design and implement an Acoustic Pattern Mining Algorithm (APMA) that analyzes stored acoustic event journal logs to surface recurring behavioral patterns and environmental insights for the user (extending EMA principles from Andersson et al., 2023; Jorgensen et al., 2025).
6. To implement a local SQLite-based acoustic event journal that records, timestamps, and stores classified sound events for longitudinal user review (Christensen et al., 2021).
7. To evaluate the performance, usability, and effectiveness of SANA through functional testing, classification accuracy benchmarking, and usability assessment among target users.

Significance of the Study

- **Individuals with Hearing Disabilities (DHH):** Directly benefits DHH individuals by providing a reliable, accessible, and on-device tool for perceiving environmental sounds, enhancing personal safety and situational awareness as called for in the literature (Jain et al., 2020; Goodman et al., 2021; Velmurugan & Prabhakar, n.d.).
- **Healthcare and Assistive Technology Practitioners:** Contributes a novel smartphone-based approach to auditory assistance, supplementing rehabilitative strategies and introducing the first DHH-specific acoustic event journaling approach—extending EMA principles (Andersson et al., 2023; Christensen et al., 2021) to a mobile DHH context.
- **Developer Community and Scholars:** Provides a reference implementation of on-device YAMNet TFLite within Flutter (Gougeh et al., 2024), and introduces the CASPA algorithm as a foundational framework for future intelligent notification systems targeting DHH users.
- **Future Researchers:** The integration of sound classification, spatiotemporal logging, and context-aware prioritization (Huang et al., 2024; Goodman et al., 2021) opens pathways for future research in adaptive assistive mobile technologies, particularly in underrepresented cultural and linguistic contexts like the Philippines.
- **Philippine DHH Community and Policy Makers:** Provides empirical usability data (via SUS) on a locally-developed DHH assistive technology, contributing to the evidence base for policy decisions on accessible technology deployment in Philippine educational and community settings.

Scope and Limitations

The purpose of this study is to develop SANA (Sound-Aware Notification Assistant), a smartphone-based application that utilizes the YAMNet model for on-device environmental sound classification with temporal and spatial logging, and context-aware alert prioritization, designed to assist individuals with hearing disabilities.

1. **Target Platform:** Compatible with Android devices only (API level 21+); not developed for iOS.
2. **On-Device Constraint:** All ML inference (YAMNet TFLite) and data storage (SQLite) run strictly on the local device; no cloud processing or server communication is required.
3. **Sound Classes:** Limited to **28 selected sound classes** grouped as:
 - *Emergency & Safety:* Fire alarm/Smoke detector, Emergency Sirens, Screaming, Glass breaking, Explosion/Gunshot.
 - *Household & Indoor:* Doorbell/Chime, Door knocking, Phone ringing/Alarm clock, Baby cry, Dog barking, Cat meowing, Water running, Microwave beep, Smoke/CO detector.

- *Urban & Traffic*: Vehicle horn/Car honking, Brake squeal, Motorcycle engine, Train whistle, Rain/Thunder.
 - *Human Interaction*: Speech/Talking, Coughing, Sneezing, Laughter.
4. **Usability Evaluation**: Limited to **30 respondents** at Cor Jesu College (15 DHH individuals + 15 hearing individuals, aged 15+, with access to an Android smartphone).
 5. **Evaluation Instruments**: User Background Information form, System Usability Scale (SUS) Questionnaire, and debriefing questions aligned with TAM constructs (Davis, 1989).
 6. **Acoustic Pattern Mining**: The APMA analyzes only events stored within the application's local SQLite database and does not access external data sources or cloud repositories.
 7. **Spatial Logging**: GPS-based temporal and spatial logging is limited to the accuracy of the device's built-in GPS hardware; indoor location accuracy may vary.

Definition of Terms

- **Acoustic Event Journal (AEJ)**: A local, SQLite-based log maintained by SANA that records timestamped entries and GPS coordinates for each classified sound event, enabling longitudinal review following EMA principles (Andersson et al., 2023; Christensen et al., 2021).
- **Acoustic Pattern Mining Algorithm (APMA)**: An original algorithmic contribution of this study that analyzes stored AEJ logs to identify recurring sound events, temporal patterns, and behavioral trends—extending data-logging insights (Jorgensen et al., 2025) to a DHH mobile context.
- **Context-Aware Sound Prioritization Algorithm (CASPA)**: An original algorithmic contribution of this study that dynamically ranks classified acoustic alerts by computing a composite priority score derived from model confidence, temporal cooldowns, time of day, environmental zone, and user-defined sound urgency—addressing notification fatigue (Jain et al., 2020) and maximizing user safety awareness.
- **Deaf and Hard-of-Hearing (DHH)**: Individuals with significant or total hearing loss who primarily use visual or sign language modalities. As documented by Goodman et al. (2021), this community exhibits diverse and individualized acoustic awareness needs.
- **Ecological Momentary Assessment (EMA)**: Repeated, real-time data collection about a participant's environment in situ (Andersson et al., 2023; Jorgensen et al., 2025). SANA applies these principles to acoustic event logging for DHH users.
- **Environmental Sound Classification (ESC)**: The computational task of automatically identifying non-speech audio signals using machine learning (Veena & Aravindhar, 2022; Mohaimenuzzaman et al., 2022).
- **Notification Fatigue**: User desensitization to system alerts due to excessive, repetitive, or poorly prioritized notifications, identified as a critical DHH adoption barrier by Jain et al. (2020).

- **System Usability Scale (SUS):** A validated 10-item Likert-scale questionnaire developed by Brooke (1996) that produces a composite usability score on a 0–100 scale; scores above 70 are considered "Good" or acceptable.
- **YAMNet:** A deep neural network architecture developed by Google that predicts audio event classes using AudioSet ontology. SANA uses a TensorFlow Lite (TFLite) version of YAMNet for on-device inference, as validated by Gougeh et al. (2024) for real-time Android sound event detection.

CHAPTER 2: REVIEW OF RELATED LITERATURE

Assistive Technology for the Deaf & Hard of Hearing

The DHH community faces compounded barriers in environmental awareness and social communication. Research in this domain has evolved from basic amplification tools toward intelligent, multi-modal, and context-aware systems.

Kim et al. (2018) developed one of the earliest systematic home monitoring assistants specifically for DHH users. Their system utilized bidirectional gated recurrent neural networks (BiGRU) for environmental sound event detection, converting detected sounds into both text display and haptic vibration patterns. The system's four-module pipeline—signal estimation, reliable sensor channel selection, sound event detection, and haptic conversion—established a foundational architecture that SANA extends by introducing contextual prioritization and longitudinal logging.

Jain et al. (2020) subsequently deployed HomeSound, an iterative in-home sound awareness system for DHH users combining multiple microphone-display units with smartwatch haptic alerts. Their three-week field deployment across four DHH homes revealed a critical limitation: while sound classification increased participants' awareness, constant smartwatch vibrations—particularly from misclassifications and repetitive events—were poorly received and led to user disengagement. This direct evidence for notification fatigue is the primary motivation for SANA's Context-Aware Sound Prioritization Algorithm (CASPA). Jain et al. (2022) later introduced ProtoSound, a personalized and scalable sound recognition system that enables DHH users to teach the system new sound classes using only a few examples via few-shot learning. Their evaluation showed that personalization substantially improved classification relevance and reduced unwanted interruptions.

Goodman et al. (2021) extended this line of inquiry by conducting a comprehensive study on user-driven sound recognizer personalization with DHH individuals, demonstrating that automated sound recognition tools are most effective when users retain control over which sounds trigger alerts and how prominently those alerts are presented. This research informs SANA's CASPA user-configurable urgency weight (U_{sound}).

Asakura (2023) explored an augmented reality-based presentation method for household sounds, displaying both machine-learning-classified sound icons and real-time spectrograms to DHH users. Their user surveys confirmed that DHH individuals strongly preferred visual representations that conveyed both the type and the temporal dynamics of surrounding sounds—supporting SANA's multi-modal visual-haptic alert design.

Chin et al. (2023) contributed a wearable edge computing device specifically for DHH road safety. Their EfficientNet-based fuzzy rank ensemble model, trained on 8,756 audio files from CREMA-D and road siren datasets, classified seven audio classes deployed on an Arduino Nano 33 BLE Sense. When a siren was detected, the device presented vibration alerts and OLED panel messages—confirming the effectiveness of haptic-visual hybrid alerting for real-world road safety scenarios. Their work justifies the inclusion of emergency siren and vehicle-related classes as high-urgency categories in SANA's sound class taxonomy.

Velmurugan & Prabhakar (n.d.) developed Vibe, an EdgeML-based sound indicator device trained on a locally collected Indian-context dataset, capable of recognizing baby cries, alarms, door knocks, vehicle horns, and spoken names. Their finding that locally trained models outperform generic pretrained models in culturally and acoustically specific environments directly motivates SANA's fine-tuning of YAMNet for Philippine acoustic conditions.

Huang et al. (2024) introduced HACS, a Human-AI Collaborative approach to sound awareness system design. HACS uses AI to identify acoustic characteristics (e.g., a "beep") and prompts DHH users to apply contextual knowledge to disambiguate the event (e.g., "microwave beep" vs. "patient monitor beep"). Their sound taxonomy—built from insights gathered through a multi-phased research process with mixed-hearing participants—provides a theoretical basis for SANA's context-based environmental zone scoring (E_{env}) in CASPA.

Fanzeres et al. (n.d.) developed an on-device mobile environmental sound recognition application for DHH users that executes all processing locally—from audio feature extraction to classification and visual result presentation. Their report of positive user feedback during field tests with deaf participants validates the user-centered design philosophy adopted in SANA.

Kashyap et al. (2025) developed a deep learning system using CNNs and RNNs for real-time audio classification aimed at enhancing environmental awareness for deaf individuals, classifying sirens, alarms, doorbells, vehicles, and speech with high accuracy and low latency—validating the feasibility of SANA's target sound class taxonomy.

Veena & Aravindhar (2022) developed SCSHIP, a DNN-based sound classification system for hearing-impaired people classifying urban sounds into categories and delivering alerts through visual and haptic modalities. Their system demonstrated that deep neural network classifiers effectively bridge the sensory gap for DHH users in urban acoustic environments—supporting SANA's dual visual-haptic alert design.

In the Philippine context, BridgeApp (n.d.) demonstrated effective two-way deaf-hearing communication via an offline mobile app supporting both American Sign Language (ASL) and Filipino Sign Language (FSL), confirming local user acceptance of digital assistive tools—a foundational baseline for SANA's local deployment and usability validation.

On-Device Environmental Sound Classification

The shift from cloud-dependent to on-device machine learning has been transformative for DHH assistive applications, enabling privacy-preserving, real-time sound classification without network dependency.

Mohaimenuzzaman et al. (2022) presented ACDNet, the first deep convolutional network specifically designed for acoustic recognition on microcontrollers with extreme resource constraints. By combining architecture design, quantization-aware training (QAT), and knowledge distillation, they achieved a 97.22% model size reduction and 97.28% FLOP reduction while maintaining close to state-of-the-art accuracy (96.25% on ESC-10, 83.65% on ESC-50). ACDNet was successfully deployed on a standard off-the-shelf microcontroller, demonstrating the technical viability of running deep audio classifiers on resource-constrained hardware—establishing the compression methodology applicable to SANA's YAMNet TFLite pipeline.

Vuegen & Karsmakers (2021) investigated CNN architectures for real-time domestic acoustic event classification on ARM Cortex-M7 microcontrollers using quantization-aware training with 8-bit fixed-point precision. Their models, trained on real-life acoustic datasets from 72 different home environments, were successfully deployed without sacrificing classification performance compared to floating-point counterparts. This study validates that mobile-grade on-device inference is achievable for DHH domestic sound monitoring.

Gougeh et al. (2024) developed ImmerseGuard, a two-stage YAMNet classification pipeline deployed on Android smartphones for safety-critical sound detection. Their system achieved 90% real-time accuracy across classes including emergency sirens, vehicles, household sounds, and device alarms—directly establishing YAMNet as a viable mobile backbone for SANA's ESC module. Their user experience study further confirmed positive system usability and user satisfaction, providing a performance benchmark for SANA.

Goulão et al. (2024) demonstrated a multi-label deep learning model for real-world deployment on street sensors, achieving 85.4% accuracy on ESC-50 via knowledge distillation pretraining. Their analysis of real-world performance gaps between benchmark datasets and deployed sensors highlights the importance of domain-specific data augmentation—a challenge SANA addresses through fine-tuning for Philippine acoustic environments.

Jekaterynczuk & Piotrowski (2024) conducted a comprehensive survey of sound source localization and detection methods, classifying approaches from propagation-based physical methods to machine learning and deep learning architectures. Their taxonomy of acoustic AI methods contextualizes SANA within the broader landscape of environmental acoustic intelligence and reinforces the selection of a neural network-based approach for SANA's classification module.

Antony et al. (2025) proposed a multi-label acoustic event detection model for automotive environments using the BEATs audio transformer and a single-layer neural network. Their mean accuracy of 0.93 across 11 sound classes on real in-car audio recordings demonstrates the generalizability of pre-trained audio encoders to domain-specific settings, providing a relevant performance reference for SANA's classification module.

Yeow et al. (2024) performed extensive real-time sound event localization and detection (SELD) deployment experiments on an edge device, identifying two critical bottlenecks: the high computational cost of feature extraction and performance degradation under low-latency inference conditions. Their findings underscore the importance of optimizing inference pipelines for real-time mobile deployment—considerations reflected in SANA's use of the TFLite quantized YAMNet model.

Context-Aware Notification and Prioritization Systems

The challenge of delivering timely, relevant, and non-fatiguing notifications has been identified as one of the most critical barriers to sustained DHH assistive technology adoption.

Jain et al. (2020) provided the foundational empirical evidence for notification fatigue in DHH assistive systems. Their HomeSound deployment demonstrated that sound classification systems without intelligent filtering mechanisms produce unacceptable levels of alert interruption, ultimately causing users to disengage from the system entirely. This finding establishes the core problem that CASPA is designed to solve.

Goodman et al. (2021) demonstrated that DHH users require control over alert prioritization to sustain system engagement. Their participatory design approach showed that users prefer systems where they can set urgency levels for individual sound categories—directly informing CASPA's user-configurable urgency parameter (U_{sound}).

Jain et al. (2022) further demonstrated through ProtoSound that user personalization of sound recognizers reduces unwanted interruptions and improves overall system acceptance. Their framework for user-driven customization is conceptually aligned with CASPA's personalization layer.

Huang et al. (2024) introduced the concept of context-dependent sound disambiguation in HACS. By incorporating environmental context (e.g., the user's current location or activity) into the sound recognition process, HACS demonstrated that context dramatically improves the precision and relevance of sound alerts. This directly informs CASPA's environmental zone scoring component (E_{env}), which uses GPS geofencing to modulate alert thresholds by context.

Jorgensen et al. (2025) characterized how acoustic environment demand and diversity vary systematically by time of day and day of week. Their finding that acoustic environments follow predictable temporal rhythms provides empirical justification for CASPA's time-of-day multiplier (T_{time}), which boosts alert priority for safety-critical sounds during high-risk periods (e.g., nighttime fire alarms).

Asakura (2023) confirmed through user surveys that DHH individuals prefer notification systems that convey temporal dynamics and contextual relevance—not merely binary alert triggers. This supports CASPA's multi-factor composite scoring approach over simpler threshold-based filtering.

Acoustic Journaling, Monitoring & Ecological Momentary Assessment (EMA)

Ecological Momentary Assessment (EMA) and acoustic data logging have been validated in audiology as methods for capturing real-world sound environment characteristics. However, their application to DHH-specific mobile software is entirely novel.

Andersson et al. (2023) investigated real-life hearing aid benefit from noise management using smartphone-based EMA combined with acoustic data logging of listening environments. Their crossover study across 11 hearing-impaired adults found that EMA combined with logged soundscape classifications provided more accurate and targeted insights than the standard SSQ12 retrospective questionnaire. Their finding that retrospective self-reports fail to capture the granularity of real acoustic environments directly motivates SANA's continuous local AEJ—eliminating recall bias for DHH users.

Christensen et al. (2021) analyzed real-world hearing aid usage patterns using remotely logged longitudinal data from 64 hearing aid users, demonstrating that smartphone-connected acoustic data logging could establish behavioral norms regarding usage patterns and associations with environmental factors. Their

framework for structuring longitudinal acoustic data directly informs the schema design of SANA's SQLite AEJ and the APMA's pattern analysis methodology.

Jorgensen et al. (2025) characterized how sound levels in listeners' acoustic environments—measured using body-worn dosimeters across three one-week sampling periods—vary systematically by time of day and day of week, but remain relatively stable across seasons. Their finding that a single week of sampling provides a statistically representative environmental profile validates SANA's design decision to surface weekly acoustic trend summaries through the APMA.

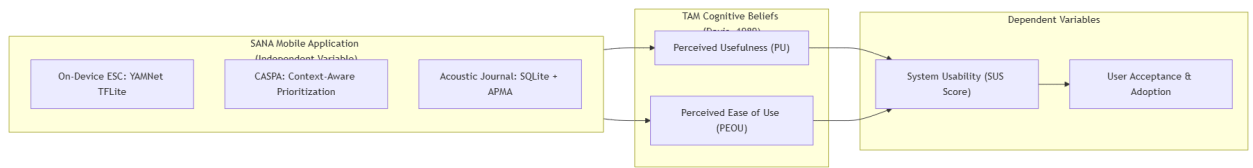
Collectively, these EMA studies establish a strong empirical precedent for the value of continuous acoustic logging and pattern analysis. SANA extends this paradigm to a DHH-specific mobile application context, where the insights derived from acoustic journaling serve not only as retrospective behavioral data but also as a tool for user empowerment and self-awareness.

The Technology Acceptance Model (TAM), formulated by Davis (1989), and the System Usability Scale (SUS) provide the evaluative framework for this study. Davis (1989) validated Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) as the primary determinants of user acceptance, with scale reliabilities of 0.98 and 0.94 respectively. The SUS operationalizes usability measurement through a 10-item scale producing scores from 0 to 100, with scores above 70 classified as "Good" (Brooke, 1996). Together, these instruments provide a rigorous and empirically validated basis for assessing SANA's usability and user acceptance.

CHAPTER 3: METHODOLOGY

Theoretical and Conceptual Framework

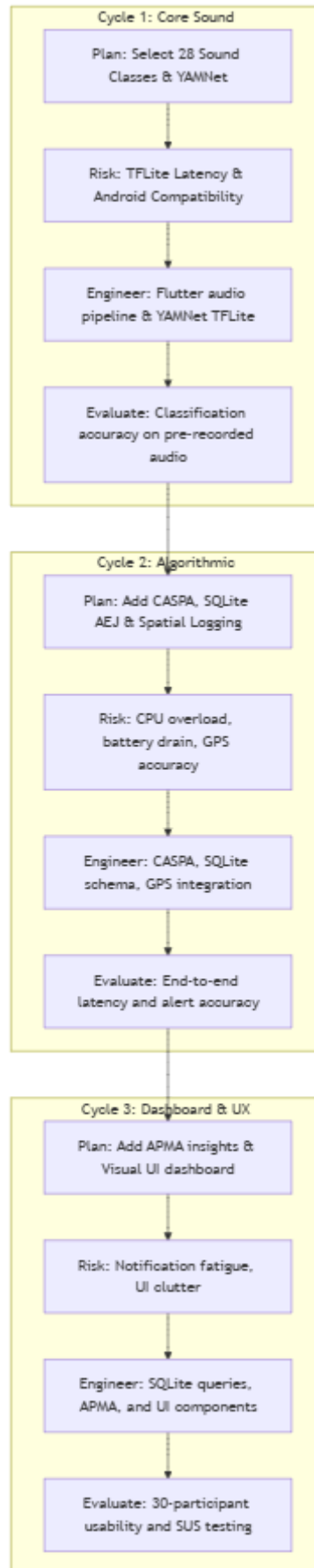
The conceptual framework of this study is anchored on the **Technology Acceptance Model (TAM)** formulated by Davis (1989). Davis validated two fundamental determinants of user acceptance: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), with scale reliabilities of 0.98 and 0.94 respectively—making TAM the most empirically robust framework for predicting IT adoption. The System Usability Scale (SUS) operationalizes PEOU measurement into a quantitative instrument.



- **Perceived Usefulness (PU):** Operationally defined as the degree to which DHH users believe SANA enhances their situational awareness, environmental safety, and daily independence (Davis, 1989). This is directly informed by the documented needs of DHH users in prior literature (Jain et al., 2020; Goodman et al., 2021; Asakura, 2023).
- **Perceived Ease of Use (PEOU):** Operationally defined as the degree to which users find the SANA interface intuitive and navigable without cognitive overload (Davis, 1989). This is measured quantitatively using the System Usability Scale (SUS).

Software Development Process: The Spiral Model (Boehm, 1988)

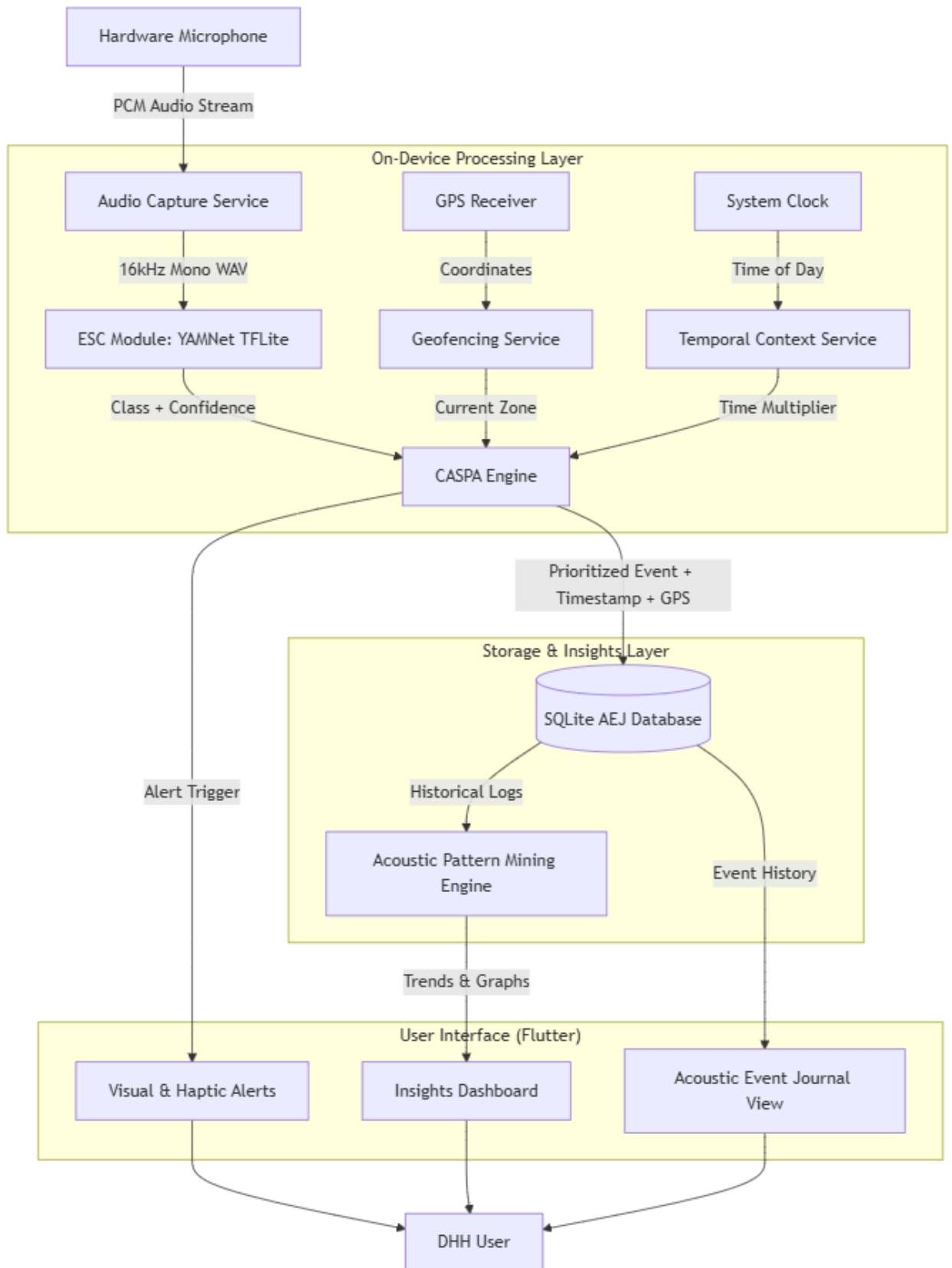
The development of SANA follows the **Spiral Model**, an iterative, risk-driven process selected because SANA integrates multiple complex modules—audio streaming, TFLite inference, local database writing, and context-aware algorithmic processing—that must coordinate in real-time under tight Android hardware constraints.



System Architecture

SANA's architecture is designed to enforce complete data privacy and on-device operability by executing all processing layers on the local device.

Logical Architecture



Physical Architecture (User Interface Components)

SANA's user interface is organized into four primary screens:

1. **Live Alert Screen:** Displays real-time sound classification results with sound category, confidence level, and CASPA priority tier (High/Medium/Low), accompanied by haptic vibration for high-priority alerts.
2. **Acoustic Event Journal Screen:** A chronological log of all classified events, displaying timestamp, GPS location, sound class, and CASPA priority score—enabling longitudinal review by the user (Christensen et al., 2021).
3. **APMA Insights Dashboard:** Visual representations of acoustic patterns derived by the APMA, including hourly distribution charts, most-frequent sound classes, anomalous event days, and sequential co-occurrence rules.
4. **Settings Screen:** Allows users to configure sound urgency weights (U_{sound}) for individual sound classes and define personal geofence zones (Home, Work, Other).

Algorithmic Formulations

1. Context-Aware Sound Prioritization Algorithm (CASPA)

Motivated by the notification fatigue documented by Jain et al. (2020) and the personalization insights of Goodman et al. (2021) and Huang et al. (2024), and informed by the temporal acoustic patterns identified by Jorgensen et al. (2025), the CASPA computes a composite priority score S_p for each classified sound event:

$$S_p = C_{class} \times R_{decay}(c) \times (W_1 \cdot U_{sound}(c) + W_2 \cdot T_{time}(t) + W_3 \cdot E_{env}(z, c))$$

Factor	Symbol	Range	Justification
Model confidence	C_{class}	[0, 1]	Suppresses uncertain predictions from generating disruptive alerts
Temporal decay cooldown	R_{decay}	[0.1, 1.0]	Prevents repeated alerts for continuous sounds (Jain et al., 2020)
User urgency score	U_{sound}	[1, 10]	User-customized ranking per sound class (Goodman et al., 2021)
Time-of-day multiplier	T_{time}	[1.0, 2.5]	Boosts priority for safety-critical sounds at high-risk hours (Jorgensen et al., 2025)
Environmental zone score	E_{env}	[1.0, 2.5]	GPS geofence zones (Home, Work, Street) — informed by HACS (Huang et al., 2024)

Decay formula: $R_{decay}(c) = 1.0 - 0.9 \cdot e^{-\lambda(t-t_{last})}$

An alert is surfaced only when $S_p \geq S_{thresh}$. When S_p exceeds the high-priority threshold, the system triggers both visual alert and haptic vibration; medium-priority events display visual-only notifications; low-priority events are silently written to the AEJ without interrupting the user.

CASPA Alert Decision Logic:

IF $S_p \geq S_{high_thresh}$ $\rightarrow$ Visual Alert + Haptic Vibration + SQLite Log
 IF $S_{thresh} \leq S_p < S_{high_thresh}$ $\rightarrow$ Visual Alert Only + SQLite Log
 IF $S_p < S_{thresh}$ $\rightarrow$ Silent SQLite Log (Background)

2. Acoustic Pattern Mining Algorithm (APMA)

Extending EMA principles (Andersson et al., 2023; Jorgensen et al., 2025) to a DHH mobile context, the APMA analyzes SQLite journal logs to surface actionable behavioral insights:

2.1 Temporal Clustering

For each sound class c , the APMA computes the hourly density distribution $D(c, h)$, where $h \in \{0, 1, \dots, 23\}$ is the hour of day:

$$D(c, h) = \frac{Count(c, h)}{\sum_{h=0}^{23} Count(c, h)}$$

This produces a temporal frequency profile for each sound class, surfaced to the user as a 24-hour bar chart on the APMA Insights Dashboard.

2.2 Anomaly Detection

For each day d and sound class c , the APMA computes the daily event count $N(c, d)$. An anomalous day is flagged when:

$$z(c, d) = \frac{N(c, d) - \mu(c)}{\sigma(c)} > 2.0$$

where $\mu(c)$ and $\sigma(c)$ are the rolling mean and standard deviation of daily counts for class c . Anomalous days are highlighted in the journal with an alert badge (Christensen et al., 2021).

2.3 Sequential Pattern Mining

The APMA identifies co-occurrence rules of the form: "Sound class A is followed by sound class B within Δt seconds in $p\%$ of occurrences," computed over the AEJ log using a sliding time window. Rules exceeding a minimum support threshold are displayed on the Insights Dashboard as behavioral patterns (e.g., "Doorbell $\rightarrow$ Dog Bark within 15s in 70% of cases").

3. SQLite Acoustic Event Journal (AEJ) Schema

The AEJ is implemented as a local SQLite database with the following schema:

Column	Data Type	Description
event_id	INTEGER (PK)	Auto-incremented unique event identifier
timestamp	DATETIME	ISO 8601 timestamp of the classified event
sound_class	TEXT	Classified sound category (one of 28 classes)
confidence	REAL	YAMNet model confidence score [0.0–1.0]
priority_score	REAL	CASPA composite score S_p
priority_tier	TEXT	HIGH / MEDIUM / LOW
latitude	REAL	GPS latitude coordinate (nullable)
longitude	REAL	GPS longitude coordinate (nullable)
geofence_zone	TEXT	Active geofence label (Home/Work/Other)
time_of_day_label	TEXT	Morning / Afternoon / Evening / Night

This schema enables both longitudinal review by the user (Objective 6) and the analytical queries required by the APMA (Objective 5), following the longitudinal data structure validated by Christensen et al. (2021).

Evaluation and Validation Plan

Automated System Testing

- **Classification Accuracy:** Precision, Recall, F1-score per class on pre-recorded datasets (ESC-50 subsets, locally recorded Philippine acoustic samples) benchmarked against Gougeh et al. (2024) (90% target accuracy).
- **Latency Benchmarking:** End-to-end classification latency from audio input to visual/haptic alert trigger, measured in milliseconds across five Android device configurations.
- **CASPA Validation:** Scenario matrices testing alert rankings across 30 simulated context combinations (3 time-of-day $\times$ 3 geofence zones $\times$ 10 priority levels), verifying expected alert suppression and escalation behaviors.
- **APMA Validation:** Functional correctness testing using synthetic AEJ datasets with known temporal patterns, verifying that the algorithm correctly identifies injected anomalies and co-occurrence sequences.

Human-Centered Usability Testing (TAM & SUS)

- **Participants:** 30 respondents (15 DHH + 15 hearing; aged 15+; Cor Jesu College).
- **Instruments:** User Background Information form, System Usability Scale (SUS, 10 items), and TAM-aligned debriefing questions (Davis, 1989).
- **SUS Scoring:**

$$\text{SUS Score} = 2.5 \times \left[\sum_{i \in \text{odd}} (x_i - 1) + \sum_{j \in \text{even}} (5 - x_j) \right]$$

- **Target Benchmark:** Mean SUS > 70 (Good/Acceptable threshold per Brooke, 1996).
- **Procedure:** Participants will be given a structured 20-minute hands-on session with SANA on a standardized Android device. The session will include: (1) a demonstration of the live alert feature, (2) review of the Acoustic Event Journal, and (3) exploration of the APMA Insights Dashboard. Participants will then complete the SUS questionnaire followed by a 10-minute TAM debriefing interview.

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